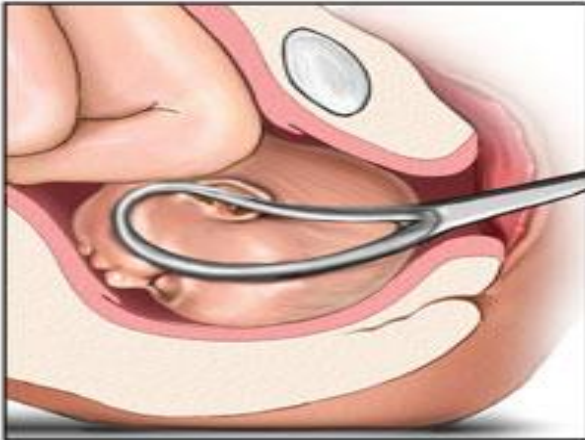


Forceps



Vacuum extraction



ASSISTED VAGINAL BIRTH

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Objective

- Understand the indications, contraindications, procedures and complications of assisted vaginal birth with vacuum or forceps.

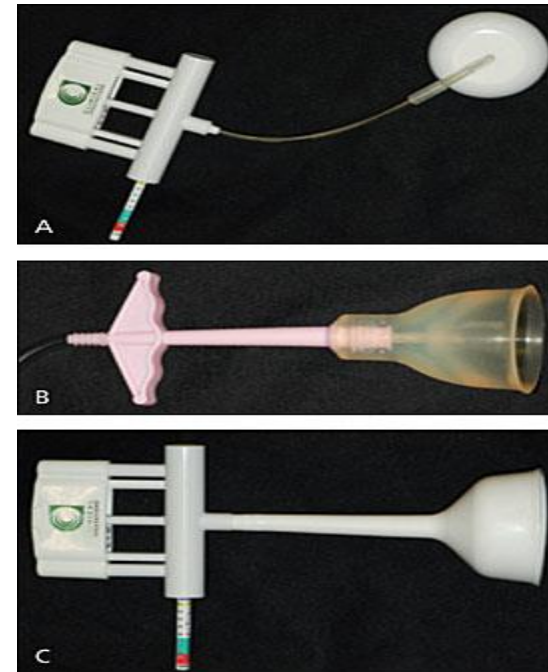
Introduction

- Assisted vaginal birth (AVB) refers to a vaginal birth with the use of any type of forceps or vacuum extractor (ventouse).
- The terms instrumental delivery, operative vaginal delivery and AVB are used interchangeably.
- The goal of AVB is to expedite the birth with a minimum of maternal or neonatal morbidity.

- As with other forms of intra-partum intervention, AVB should be performed only when the safety criteria have been met and when the benefits outweigh the risks.
- AVBs should be conducted by obstetricians with competency in the chosen procedure or by trainees under direct supervision of an experienced trainer. When these conditions are adhered to, the outcomes of AVB are good.

Incidence:

- In the UK, between 10% and 15% of deliveries are assisted with forceps or ventouse.



Indications

Fetal

- Suspected fetal compromise (pathological cardiotocograph, abnormal pH or lactate on fetal blood sampling, meconium)
- Fetal and maternal indications for assisted vaginal birth often coexist

Indications

- Maternal
- Prolonged second stage
 - Nulliparous women: lack of continuing progress for 3 hours (total of active and passive second stage of labour) with regional anaesthesia or 2 hours without regional anaesthesia
 - Multiparous women: lack of continuing progress for 2 hours (total of active and passive second stage of labour) with regional anaesthesia or 1 hour without regional anaesthesia
- Maternal exhaustion/distress
- Medical indications to avoid prolonged pushing or Valsalva (e.g. cardiac disease, hypertensive crisis, cerebral vascular disease, uncorrected cerebral vascular malformations, myasthenia gravis, spinal cord injury)
- Combined

Table 13.3 Safety criteria for assisted vaginal birth

Area	Criteria
Full abdominal and vaginal examination	Head is fully engaged (zero fifths palpable) or no more than one-fifth palpable per abdomen
	Cervix is fully dilated and the membranes ruptured
	Station at level of ischial spines or below (0/+1/+2/+3)
	Exact position of the head has been determined so correct placement of the instrument can be achieved
	Caput and moulding is no more than moderate (+ or ++)
	Pelvis is deemed adequate
Preparation of the mother	Clear explanation given and informed consent obtained
	Trust has been established and the woman offers full cooperation
	Appropriate anaesthesia is in place; for mid-pelvic rotational delivery this will usually be a regional block; a pudendal block may be appropriate, in the context of urgency; a perineal block may be sufficient for low-pelvic or outlet delivery
	Maternal bladder has been emptied recently
	In-dwelling catheter has been removed or balloon deflated
	Aseptic technique
Preparation of staff	Operator has the knowledge, experience and skill necessary
	Adequate facilities are available (appropriate equipment, bed, lighting) and access to an operating theatre
	Back-up plan in place in case of failure to deliver. For mid-pelvic deliveries, theatre staff should be available immediately to allow a caesarean section to be performed without delay (<30 minutes); senior obstetrician should be present if a junior obstetrician is conducting the delivery
	Anticipation of complications that may arise (e.g. shoulder dystocia, post-partum haemorrhage)
	Personnel present that are trained in neonatal resuscitation

Source: Adapted from RCOG Guidelines 2000

Criteria

Full abdominal and vaginal examination :

- 1- Head is fully **engaged** (zero fifths palpable) or no more than one-fifth palpable per abdomen
- 2- Cervix is **fully dilated** and the membranes ruptured
- 3- **Station** at level of ischial spines or below (0/+1/+2/+3)
- 4- Exact **position** of the head has been determined so correct placement of the instrument can be achieved
- 5- Caput and **moulding** is no more than moderate (+ or ++)
- 6- Pelvis is deemed **adequate**

criteria

Preparation of the mother:

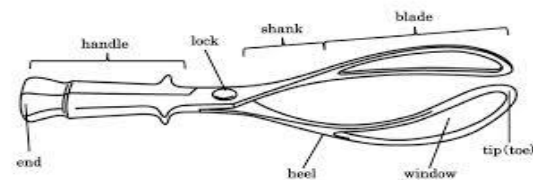
- 1- Clear explanation given and **informed consent** obtained
- 2- Appropriate **anaesthesia** is in place; for mid-pelvic rotational delivery this will usually be a regional block; a pudendal block may be appropriate, in the context of urgency; a perineal block may be sufficient for low-pelvic or outlet delivery
- 3- Maternal **bladder** has been **emptied** recently In-dwelling catheter has been removed or balloon deflated
- 4- **Aseptic** technique

criteria

Preparation of staff :

- 1- Operator has the knowledge, **experience** and skill necessary
- 2- **Adequate facilities are available** (appropriate equipment, bed, lighting) and access to an operating theatre
- 3- **Back-up plan in place in** case of failure to deliver. senior obstetrician should be present if a junior obstetrician is conducting the delivery
- 4- **Anticipation of complications** that may arise (e.g. shoulder dystocia, post-partum haemorrhage)
- 5- Personnel present that are trained in **neonatal resuscitation**

- For forceps, all of the usual prerequisites for safe AVB apply, but, in addition, it is essential that the **operator checks the pair of forceps to ensure that a matching pair** has been provided and that the blades lock with ease (both before and after application).



ASSESSMENT

- A thorough **abdominal** and **vaginal** examination should take place to confirm the fetal lie, presentation, engagement, station, position, attitude and degree of caput or moulding.

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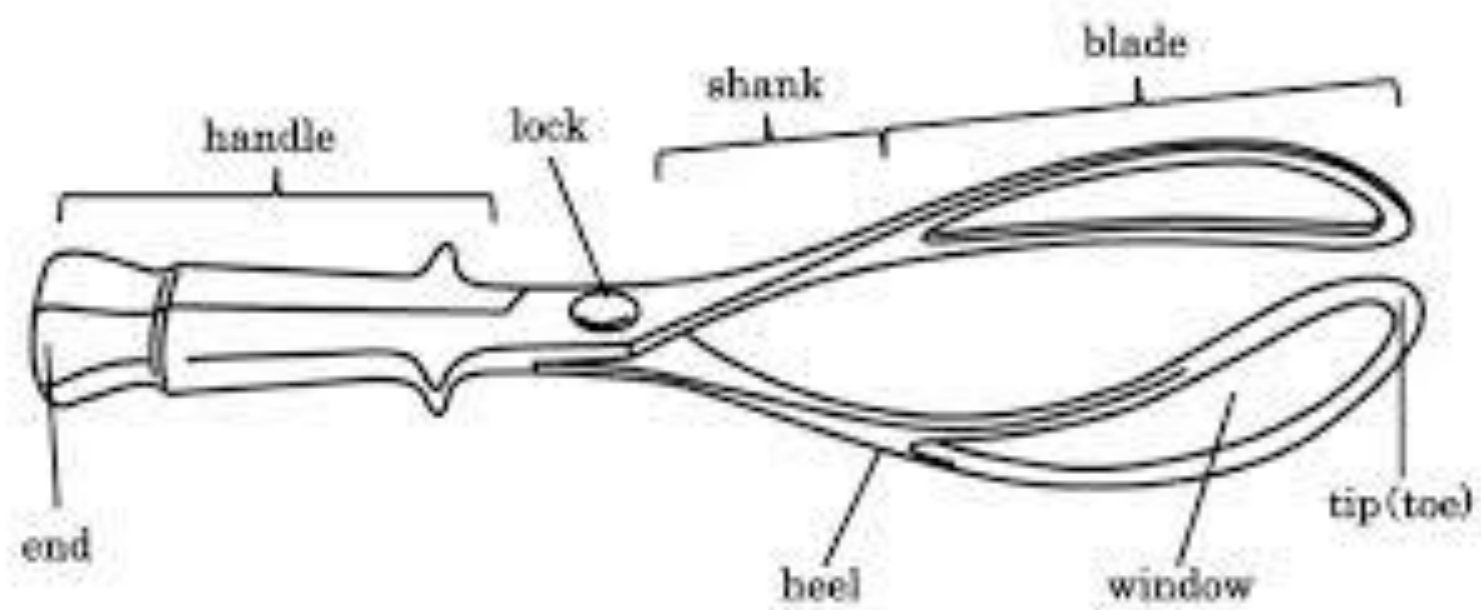
- A pelvic examination will determine whether there are **any 'mechanical' contraindications** to performing an AVB.
- If, for example, a contracted pelvis is the cause of failure to progress in the second stage, then consideration should be given to the choice of instrument or whether it may be more prudent to perform a caesarean section.
- The angle of the subpubic arch, the curve of the sacral hollow and the presence of prominent ischial spines all contribute to the decision of whether an AVB may be safely performed.

CONTRAINDICATIONS

- When the safety criteria are not met, AVB is contraindicated, for example a high fetal head two-fifths palpable abdominally with station above the ischial spines.
- The vacuum should not be used in gestations of less than 34 completed weeks because of the risk of cephalohaematoma and intra- cranial haemorrhage.
- It should not be used for a face or breech presentation.
- There is minimal additional risk of fetal haemorrhage if the vacuum extractor is employed following fetal blood sampling or application of a fetal scalp electrode.
- Forceps and vacuum extractor deliveries before full dilatation of the cervix are contraindicated.

Forceps delivery:

- Most obstetric forceps have a pair of fenestrated blades with a cephalic and pelvic curve between the heel and toes.
- The cephalic curve permits the instrument to be applied to the sides of fetal head, and the pelvic curve corresponds to the curved axis of the maternal pelvis. The tip of blade called toe while the other end called heel. The heel continues as a shank which ends in the handle.



Forceps delivery:

- The cephalic curve is constructed to grasp the fetal head, with the toes of the blades over the maxilla or malar eminences while the length of the blade grasps the sides of the head from the malar area along the sides of the head in front of the ear and the parietal bones in front of the occiput.
- This bimalar biparietal application exerts uniform pressure on the head. In this position the shank is over the flexion point and thus allows the correct direction of traction

Functions:

- **Traction**: the most important, it requires 18kg in primigravida and 13kg in multipara.
- **Compression**: should be minimal when properly applied.
- **Rotation**: this is when Keillands forceps is used.
- **Protection**: as in preterm.

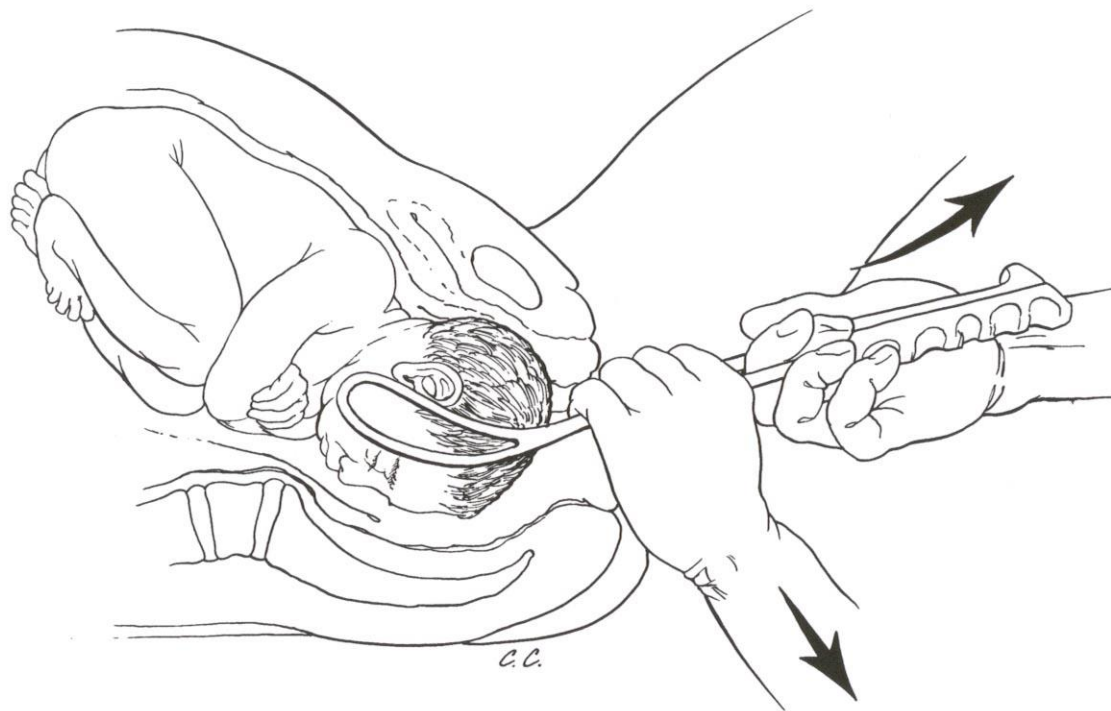
Technique:

Check that both pairs are matching. By convention, the left blade is inserted before the right hand which use in guiding blade and protecting the vaginal wall. The right blade is inserted next. With proper placement, they come to lie parallel to the axis of fetal head and between fetal head and pelvic wall.



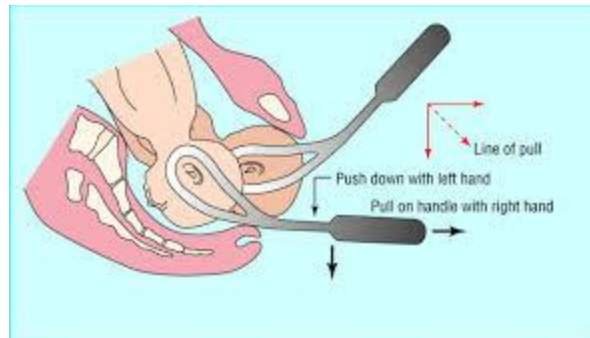
Technique:

- The blades should articulate with ease indicating correct application
- Traction should be applied intermittently in concert with uterine contraction and maternal expulsive efforts. The axis of traction changes during the delivery and is guided along the J-shaped curve of the pelvis i.e. first horizontal, backward, forward and lastly upward.



Technique:

- As the head begins to crown, the blades are directed to the vertical and an episiotomy is performed. Once the head is delivered, the head are removed in the reverse order to which they were inserted.
- **In face presentation**; blades are applied along occiputomenal diameter.
- **In after-coming head** Pipers forceps, the blades are applied from below after elevating the legs.



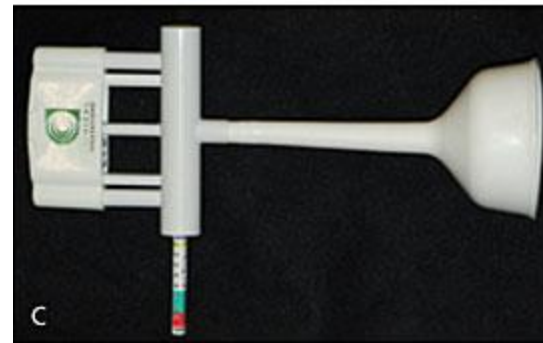
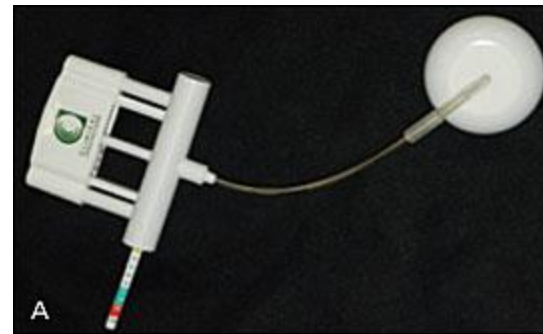
Ventouse /vacuum extractors:

- The basic component of this instrument is that a **suction cup**, of a **silastic** or **rigid** construction, is connected, via **tubing**, to a **vacuum source**. Either directly through the tubing or via a connecting chain, direct traction can then be applied to the presenting part to expedite delivery.

- The cups comes in different sizes and are usually 4, 5 or 6 cm in diameter. Soft plastic cup where the tubing comes from the center of the cup is suitable when the head is occipitoanterior. For the occipitoposterior position the tube should comes from lateral side and rigid plastic or metal cup is preferred.

Indications:

Is the same as the forceps.



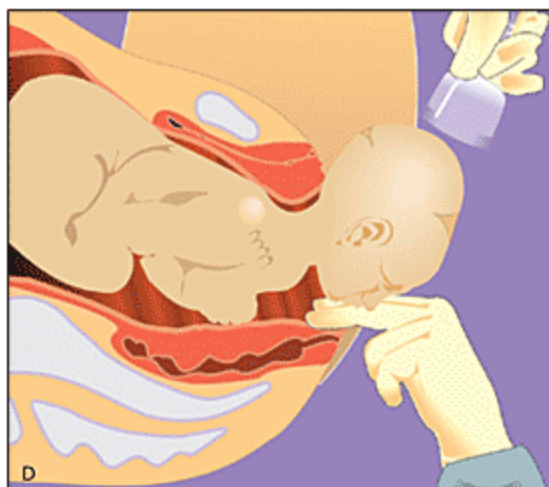
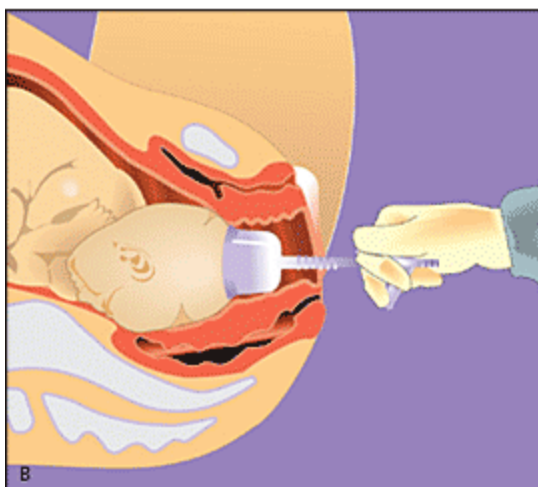
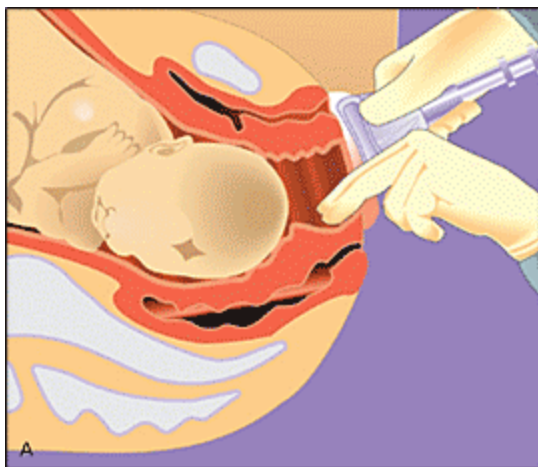
Functions:

- Suction: by effect of negative pressure which lead to the formation of caput or chignon.
- Traction: in the direction of pelvic axis.

Technique:

For successful application, determination of flexion point is vital. This is on the sagittal suture 3 cm anterior to posterior fontanelle and 6cm posterior to anterior fontanelle. The center of the cup should be over this point. in majority of the cases a cup size of 5 cm usually used .

- The pressure used for nearly all vacuum delivery is between -0.6 and -0.8 kg/cm² (-60 to -80 kPa).
- Traction must occur in the plane of least resistance along the axis of pelvis. Safe and gentle traction is then applied in concert with uterine contraction and voluntary maternal expulsive efforts.

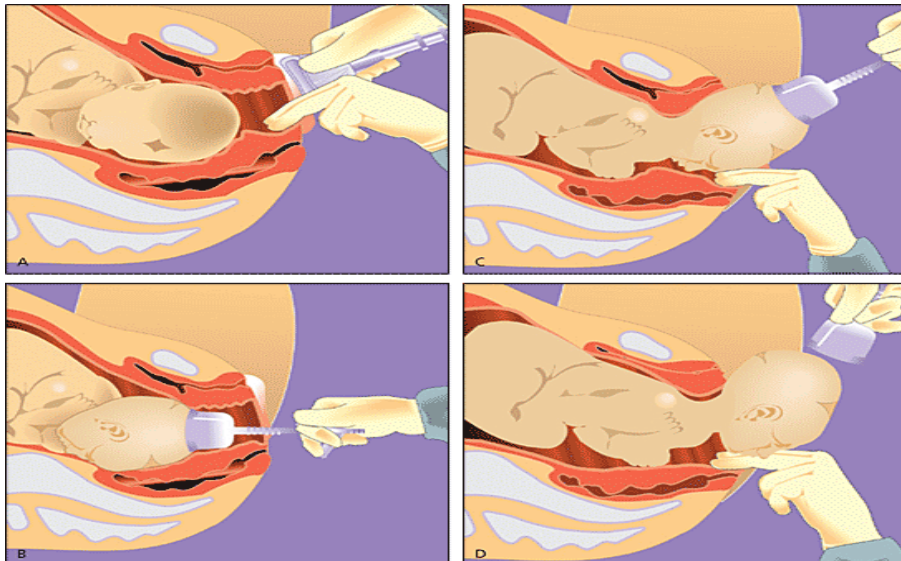


CHOICE OF INSTRUMENT

- The choice of instrument should be based on a combination of **indication**, **experience** and **training**.
- The aim should be to complete the delivery successfully with the lowest possible morbidity, and the preferences of the mother should be taken into account.

CHOICE OF INSTRUMENT

- Vacuum is preferred as a first-line instrument by many obstetricians in terms of reduced maternal trauma, but this needs to be balanced with a failure rate of 10–20% compared with a failure rate with forceps of 5% or less for similar deliveries.



- The morbidities for the baby differ, with a higher incidence of cephalohaematoma and cerebral haemorrhage with vacuum and a higher incidence of lacerations and facial palsy with forceps.

CHOICE OF INSTRUMENT

Vacuum (compared with forceps) is significantly **more** likely to be associated with:

- failure to achieve a vaginal birth
- cephalohaematoma (subperiosteal bleed)
- retinal haemorrhage
- maternal worries about the baby

Vacuum (compared with forceps) is significantly **less** likely to be associated with:

- the use of maternal regional/general anaesthesia
- significant maternal perineal and vaginal trauma
- severe perineal pain at 24 hours
- Vacuum and forceps are similar in terms of:
- delivery by caesarean section (where failed vacuum is completed by forceps)
- low 5-minute Apgar scores

ANALGESIA

- Analgesic requirements are greater for forceps than for vacuum-assisted births.
- When **rotational forceps** or **mid-pelvic direct traction** forceps are needed, **regional analgesia** is preferred.
- For **a rigid cup vacuum-assisted** delivery, a **pudendal block** with perineal infiltration may be all that is needed and, if a soft cup is used, analgesic requirements may be limited to perineal infiltration with local anaesthetic.

POSITIONING

- AVBs are traditionally performed with the patient in the **lithotomy position**.
- In patients with limited abduction (such as those with symphysis pubis dysfunction), it may be necessary to limit abduction of the thighs to a minimum.
- The bladder **should be emptied** with a catheter to avoid injury.

CONTINGENCY PLANNING

- With any AVB, there is the potential for failure with the chosen instrument and the operator must have a back-up plan for such an event.
- It may be possible to **complete a failed vacuum-assisted birth with low-pelvic forceps,**
- but failed forceps delivery will **almost always result in caesarean section.**

- With any difficult AVB, the risk of **shoulder dystocia** after successful delivery of the fetal head should be considered, as should the potential for **post-partum haemorrhage**.
- The operator must develop the **skills** necessary to anticipate such events and to manage the consequences
- . If the AVB is being performed for suspected fetal compromise, then the need for **neonatal resuscitation** should be anticipated and a neonatologist should have been called to attend.

- The crowning phase should occur shortly afterwards and, depending on the resistance of the perineum, may occur with one further pull or some operators prefer to use up to three very small pulls to minimize perineal trauma.

-

- With any **vacuum**, the operator should allow **no more than two episodes** of breaking the suction 'pop-offs' in a vacuum-assisted birth, and the maximum time from application to delivery should ideally be **less than 15 minutes**.
- Rotation is achieved by the natural progression of the head through the pelvis.

- It is not acceptable to use a vacuum when the **position** of the fetal head is **unknown**, there is a **significant degree of caput** that may either preclude correct placement of the cup or, more sinisterly, indicate a substantial degree of cephalopelvic disproportion, or the **operator is inexperienced** in the use of the instru- ment.

- Forceps-assisted births should be completed in **no more than three pulls**.
- Specific techniques are required for rotational forceps deliveries and only those who have been properly trained in their use should employ them.
- Rotation occurs between contractions and the descent phase is similar to non- rotational forceps.

EPISIOTOMY

- The role of episiotomy at vacuum and forceps delivery is **controversial**, with conflicting studies reported.
- In practice, most obstetricians cut an episiotomy routinely for forceps-assisted delivery especially in nulliparous births where anal sphincter damage is more likely.
- In parous women, particularly those requiring vacuum-assisted birth, an episiotomy may not be necessary..

SPECIAL CONSIDERATIONS

- **FAILURE OF THE CHOSEN INSTRUMENT**
- Failure to complete delivery vaginally can occur when
 - the choice of instrument is wrong,
 - the application of the instrument is wrong) or
 - when the position has been wrongly defined
- Failure is also more common if the fetus is large or maternal effort is poor.

- When the first instrument fails, a number of scenarios may apply.
- If the reason for failure was cup detachment of a vacuum and the fetal head is occipito-anterior and on the perineum, low-pelvic or lift-out forceps to complete delivery is acceptable and likely to be less traumatic than a second-stage caesarean section.

- If the instrument failed because there was little or no descent with the first pull of a correctly applied instrument with traction in the correct axis of the pelvis, then delivery must be by caesarean section, as the likely diagnosis is cephalo-pelvic disproportion.

-

- If the instrument failed because the position was incorrectly defined, then the next option will be either a rotational instrumental delivery or a caesarean section.
- If there is any uncertainty, senior help should be sought immediately and a full re-evaluation should take place, ideally in an operating theatre. In many cases, delivery by caesarean section will be the safer option for the fetus.

Trial of IVD

- Its indicated when it difficult to judge the success of IVD.
- It should be done in OR under good epidural or spinal anesthesia and with the theater team, anesthetist and pediatrician present.

Complications:

Maternal:

1. Need for analgesia.
2. Post partum hemorrhage due to uterine atony or trauma.
3. Higher rate of genital tract trauma including 3rd and 4th degree tear with its subsequent sequel like fecal incontinence.
4. Extension of episiotomy with increased blood loss.
5. Infections due to improper aseptic technique.

Complications:

Maternal:

Fetal:

1. Facial bruising and trauma.
2. Facial or brachial nerve palsy.
3. Cephal hematoma.
4. Intra cranial hemorrhage.
5. Skull fracture.
6. Asphyxia and death.

- **KEY LEARNING POINTS**

- ~ AVB should be classified according to the position and station of the presenting part.
- ~ Clinical assessment and confirmation that the safety criteria have been met are essential prior to AVB.
- ~ Vacuum or forceps may be suitable depending on the clinical circumstances and operator's preference.
- ~ AVBs with a higher chance of failure should be conducted in an operating theatre.
- ~ Contingency planning is an essential part of any AVB.
- ~ Anticipation and early management of maternal and neonatal complications is essential.

THANK YOU